



DWF_dp_mult_sat functions

Multiply and Saturate

Version, STAR, and myDesignWare Subscriptions: [IP Directory](#)

Description

Revision History

The DWF_dp_mult_sat function multiplies the two arguments *a* and *b*, truncates the upper bits of the result to the width specified by argument *p_width* and returns a saturated value if an overflow (or underflow) occurs. A dedicated overflow detection is used to improve QoR of the multiplier.

Table 1-1 Function Names

Function Name	Description
DWF_dp_mult_sat	VHDL unsigned multiply and saturate
DWF_dp_mult_sat	VHDL signed (two's complement) multiply and saturate
DWF_dp_mult_sat_uns	Verilog unsigned multiply and saturate
DWF_dp_mult_sat_tc	Verilog signed (two's complement) multiply and saturate

Table 1-2 Argument Description

Argument Name	Type	Width / Values	Description
<i>a</i>	Vector	<i>a_width</i>	Input multiplier
<i>b</i>	Vector	<i>b_width</i>	Input multiplicand
<i>p_width</i>	Integer	≥ 2	Word length of returned value (VHDL only, constant)
DWF_dp_mult_sat	Vector	<i>p_width</i>	Returned value

Table 1-3 Parameter Description (Verilog)

Parameter	Values	Description
<i>a_width</i>	≥ 2	Word length of input <i>a</i>
<i>b_width</i>	≥ 2	Word length of input <i>b</i>
<i>p_width</i>	≥ 2	Word length of returned value

Verilog Include File: DW_dp_mult_sat_function.inc

Functional Description

$z[p_width-1:0] = \text{DWF_dp_mult_sat}(a[a_width-1:0], b[b_width-1:0], p_width)$

Unsigned Multiply and Saturate

$$\begin{aligned} p[a_width+b_width-1:0] &= a[a_width-1:0] * b[b_width-1:0] \\ z[p_width-1:0] &= 2^{p_width-1} \quad \text{if } p[a_width+b_width-1:0] > 2^{p_width-1} \\ &= p[p_width-1:0] \quad \text{else} \end{aligned}$$

Signed Multiply and Saturate

$$\begin{aligned} p[a_width+b_width-1:0] &= a[a_width-1:0] * b[b_width-1:0] \\ z[p_width-1:0] &= 2^{p_width-1}-1 \quad \text{if } p[a_width+b_width-1:0] > 2^{p_width-1}-1 \\ &= -2^{p_width-1} \quad \text{else if } p[a_width+b_width-1:0] < -2^{p_width-1} \\ &= p[p_width-1:0] \quad \text{else} \end{aligned}$$

For more information about the DesignWare datapath functions, refer to the topic titled [DesignWare Datapath Functions Overview](#).

Related Topics

- [DesignWare Datapath Functions Overview](#)
- [DesignWare Building Block IP User Guide](#)

VHDL Example

```
library IEEE, DWARE;
use IEEE.std_logic_1164.all;
use IEEE.numeric_std.all;
use DWARE.DW_dp_functions.all;
-- DWARE.DW_dp_functions_arith package if IEEE.std_logic_arith is used

entity DWF_dp_mult_sat_test is
  port (a, b, c : in  signed(7 downto 0);
        z       : out signed(7 downto 0));
end DWF_dp_mult_sat_test;

architecture rtl of DWF_dp_mult_sat_test is
begin
  z <= DWF_dp_mult_sat (a, b, 8) + c;
end rtl;
```

Verilog Example

```
module DWF_dp_mult_sat_test (a, b, c, z);

  input  signed [7:0] a, b, c;
  output signed [7:0] z;

  // Passes the parameters to the function
  parameter a_width = 8;
  parameter b_width = 8;
  parameter p_width = 8;

  // add "$SYNOPSIS/dw/sim_ver" to the search path for simulation
  `include "DW_dp_mult_sat_function.inc"

  assign z = DWF_dp_mult_sat_tc (a, b) + c;

endmodule
```

Revision History

For notes about this release, see the [DesignWare Building Block IP Release Notes](#).

For lists of both known and fixed issues for this component, refer to the [STAR report](#).

Date	Release	Updates
September 2024	DWBB_202409.0	■ Updated date and version; no technical changes
December 2023	DWBB_202312.0	■ Updated date and version; no technical changes ■ Added this Revision History table and the document links on this page

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